

STE-CLOTILDE, QC, Canada

WMO: 716140

Lat: 45.1667N

Lon: 73.6783W

Elev: 53

StdP: 100.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-21.9	-29.2	0.3	-24.0	-26.4	0.3	-21.4	9.5	-1.3	8.4	-5.7	2.1	270	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	30.4	21.9	28.9	21.2	27.4	20.4	23.7	28.3	22.6	26.8	21.7	25.6	3.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	17.0	26.1	21.2	16.0	25.0	20.3	15.1	24.0	71.2	28.2	67.1	27.0	63.5	25.6	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.9	6.1	DB	-29.9	33.0	4.1	1.4	-32.9	34.0	-35.3	34.8	-37.6	35.6	-40.5	36.6	(4)
(5)			WB	-30.1	25.4	4.0	1.0	-33.0	26.1	-35.4	26.7	-37.6	27.3	-40.6	28.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-9.0	-7.4	-1.8	6.0	13.2	18.2	20.6	19.5	15.4	8.6	2.2	-4.5
	DBStd	11.51	7.57	6.65	6.24	4.74	4.26	3.81	2.90	3.04	4.17	4.51	5.22	6.42
	HDD10.0	2371	591	486	368	137	16	0	0	0	4	82	238	449
	HDD18.3	4412	849	719	625	370	168	48	10	23	107	302	484	707
	CDD10.0	1213	0	0	2	18	114	246	328	295	166	39	4	1
	CDD18.3	212	0	0	0	1	9	44	81	59	18	1	0	0
	CDH23.3	1905	0	0	1	16	121	420	675	488	175	9	0	0
	CDH26.7	472	0	0	0	3	29	114	169	116	40	1	0	0
Wind	WSAvg	2.6	3.1	3.1	3.1	3.2	2.7	2.4	2.1	1.9	2.0	2.5	2.7	2.9
Precipitation	PrecAvg	960	62	54	58	80	85	95	100	95	89	94	76	79
	PrecMax	1117	112	99	101	142	145	179	146	176	179	192	168	152
	PrecMin	819	26	20	18	13	28	38	53	29	31	32	21	28
	PrecStd	98	23	19	20	34	36	36	27	34	32	42	28	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	9.9	17.6	25.6	29.9	32.0	32.5	31.9	30.8	24.6	18.1	11.8
		MCWB	8.9	6.7	12.8	16.0	19.1	22.8	23.9	22.9	22.1	18.1	14.3	10.0
	2%	DB	6.3	6.2	12.2	20.8	26.7	29.8	30.1	29.4	27.3	21.4	15.0	8.4
		MCWB	4.6	3.9	8.0	13.0	17.7	21.1	22.5	21.8	20.7	16.1	12.3	6.6
	5%	DB	3.9	4.1	8.8	17.3	24.0	27.7	28.5	27.7	25.3	18.5	12.3	5.9
		MCWB	2.6	2.1	5.6	10.9	15.8	20.2	21.6	20.8	19.5	14.4	9.8	4.5
	10%	DB	1.7	2.3	6.4	14.3	21.6	25.7	27.0	26.1	23.1	16.1	10.1	3.7
		MCWB	0.6	0.9	3.8	8.9	15.1	19.0	20.7	20.2	18.2	13.1	7.8	2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	6.8	13.5	17.7	21.2	24.4	25.2	24.3	23.4	19.7	15.1	10.3
		MCDB	9.7	8.5	16.8	23.6	27.1	29.6	30.1	28.8	28.2	23.1	17.0	11.5
	2%	WB	4.9	4.1	8.6	14.4	19.4	22.8	24.0	23.0	21.7	16.9	12.6	6.9
		MCDB	5.7	5.9	11.4	19.5	24.0	27.6	28.4	27.7	25.6	20.1	14.8	8.1
	5%	WB	2.6	2.4	6.0	11.6	17.7	21.5	22.8	22.0	20.4	15.0	10.2	4.6
		MCDB	3.7	3.7	8.4	16.0	21.9	25.6	26.6	25.8	24.2	18.2	12.0	5.9
	10%	WB	0.6	1.0	4.0	9.5	16.0	20.2	21.8	21.1	19.0	13.2	7.9	2.6
		MCDB	1.6	2.2	6.3	13.4	20.3	24.1	25.5	24.7	22.4	15.5	9.9	3.6
Mean Daily Temperature Range	5% DB	MDBR	9.6	9.8	9.5	10.4	11.6	11.0	10.8	11.2	11.7	9.8	8.2	7.9
		MCDBR	9.9	10.4	12.7	16.4	15.9	13.8	12.8	13.2	14.1	13.7	11.9	9.2
		MCWBR	9.3	8.8	9.2	9.8	8.1	7.0	6.6	6.8	8.2	8.9	9.5	8.1
	5% WB	MCDBR	9.6	9.5	11.9	14.5	12.2	11.6	10.9	11.5	12.3	12.1	11.4	8.8
		MCWBR	9.1	8.4	9.1	9.6	7.5	6.7	6.2	6.5	7.6	8.6	9.7	8.1
Clear-Sky Solar Irradiance	taub	0.269	0.290	0.326	0.363	0.387	0.446	0.435	0.417	0.375	0.336	0.309	0.294	
	taud	2.369	2.304	2.348	2.358	2.347	2.187	2.224	2.270	2.376	2.490	2.490	2.345	
	Ebn at Noon	851	894	906	897	882	828	831	833	842	834	799	771	
	Edh at Noon	74	96	107	116	121	143	136	125	104	81	67	70	
All-Sky Solar Radiation	RadAvg	1.44	2.35	3.56	4.36	5.10	5.56	5.63	5.00	3.99	2.31	1.42	1.02	
	RadStd	0.16	0.24	0.28	0.51	0.51	0.44	0.43	0.23	0.33	0.22	0.17	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+62	N/A

Nomenclature: See separate page